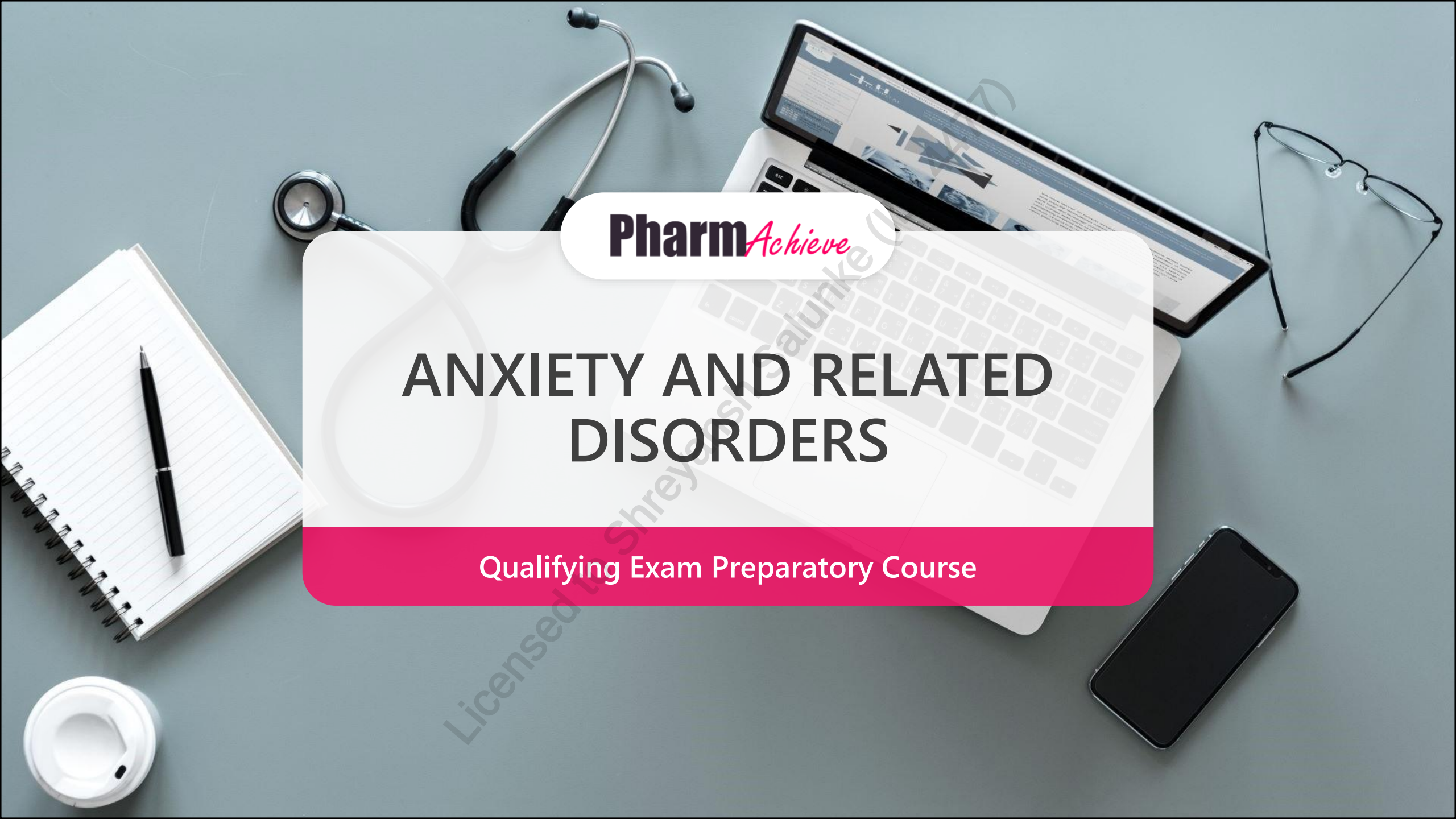




Pharm*Achieve*

ANXIETY AND RELATED DISORDERS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

CAD	Coronary Artery Disease	PNS	Parasympathetic Nervous System
CBT	Cognitive Behavioural Therapy	PRN	As needed
CBC	Complete Blood Count	PTSD	Post-traumatic Stress Disorder
COPD	Chronic Obstructive Pulmonary Disease	SAD	Social Anxiety Disorder
CVD	Cardiovascular Disease	SNRI	Serotonin Norepinephrine Reuptake Inhibitor
GABA	Gamma-amino Butyric Acid	SNS	Sympathetic Nervous System
GAD	Generalized Anxiety Disorder	SOB	Shortness Of Breath
Hx	History	SUD	Substance Use Disorder
MAOIs	Monoamine Oxidase Inhibitors	SSRI	Selective Serotonin Reuptake Inhibitor
NE	Norepinephrine	TCA	Tricyclic Antidepressants
OCD	Obsessive Compulsive Disorder	TID	Three Times Daily
PD	Panic Disorder	TSH	Thyroid Stimulating Hormone

PATHOPHYSIOLOGY - ANXIETY

Anxiety can be caused by **abnormal function** in:

Amygdala

Assesses fear stimuli and learned response to fear



Locus ceruleus

Primary NE containing site



Hippocampus

Automates responses to threats

Noradrenergic Theory

- Overreaction in response to fear
- **Overactivation of NE, SNS, PNS**

GABA Receptor and Serotonin Theory

- Not well understood

TYPES OF ANXIETY AND RELATED DISORDERS

Anxiety is a normal response to stress; resulting in anticipation, interpretation, and recollection of stressors

- An anxiety disorder is diagnosed when an individual experiences:
 - Persistent, severe anxiety symptoms and possesses irrational fears that significantly impair normal daily functioning

OCD and PTSD are not considered anxiety disorders

Generalized anxiety disorder	Social anxiety disorder	Panic disorder	Obsessive compulsive disorder	Post-traumatic stress disorder
Excessive worry that is hard to control even when there is no reason for concern on more days than not over ≥6 months	Intense anxiety provoked by social situations where embarrassment may occur; often leads to avoidance	Recurring panic attacks with persistent anxiety concerning reoccurrence; may result in avoidance	<u>Obsessions</u> : recurring intrusive thoughts, images, or urges that provoke anxiety and are unwanted/repetitive and/or difficult to control <u>Compulsions</u> : repetitive behaviours to decrease anxiety Generally, starts in childhood/adolescence	Significant distress or impairment in functioning in response to a traumatic event; lasting at least 1 month

CLINICAL PRESENTATION - ANXIETY

Psychological and Cognitive Symptoms

- Excessive **anxiety**
- Worries which are difficult to control
- Feeling keyed-up or on edge
- **Poor concentration** or mind going blank

Physical Symptoms

- Restlessness
- Fatigue
- Muscle tension
- Sleep disturbance
- Irritability

Impairment

- Social, occupational or other important functional areas
- Poor coping abilities
- Persistent and excessive signs and symptoms for **≥6 months**

DIAGNOSIS

The **Diagnostic and Statistical Manual of Mental Disorders 5 (DSM-5)** is used by healthcare providers to **assess** and **diagnose** anxiety disorders.

SCREENING AND MONITORING TOOLS

GAD-7	• Generalized Anxiety Disorder – 7 item • 7 anxiety symptoms where individuals self-report frequency of the symptoms from not at all to nearly every day • Score is used to help determine the severity of anxiety
HAM-A	• Hamilton Anxiety Rating Scale – A • Used by clinicians to rank the severity of anxiety symptoms • Broken down into 14 categories • Not as frequently used, as some guidelines state it measures depressive symptoms in addition to anxiety symptoms

RISK FACTORS AND CAUSES - ANXIETY

Risk Factors

- Family history
- Past personal history (including shyness as a child)
- Stressful childhood life events or trauma
- Poverty, unemployment
- Isolation (poor social support)
- Comorbidity with other psychiatric diagnosis and substance abuse
- Chronic physical illnesses (e.g. CVD, diabetes)
- Female gender (women 16% vs men 8%)

Medication Causes *Use or Discontinuation*

- Albuterol, theophylline, digoxin
- Anticholinergics, antihistamines
- Carbamazepine
- Stimulants & depressants
- Prednisone
- Pseudoephedrine
- Herbals: Ma Huang, ginseng, ephedra
- Levodopa
- Levothyroxine
- Quinolone, isoniazid
- SSRIs, TCAs

Medical Causes

- **Cardiovascular:** angina, heart failure, arrhythmia
- **Endocrine & metabolic:** hyperparathyroidism, hyperthyroidism, hypoglycemia
- **Neurologic:** poor pain control
- **Respiratory:** asthma, COPD
- Vitamin B12 deficiency
- Encephalitis

GOALS OF THERAPY - ANXIETY

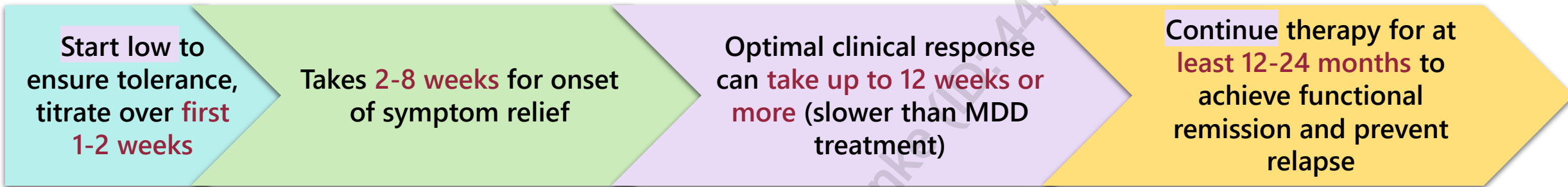
- ✓ Eliminate or decrease symptomatic anxiety
- ✓ Eliminate or decrease anxiety-based disability
- ✓ Prevent recurrence/relapse
- ✓ Achieve complete remission and recover
- ✓ Treat co-morbid conditions (e.g. depression, mood disorders)

NON-PHARMACOLOGICAL MEASURES

- CBT is 1st-line!
- Aerobic exercise
- Good sleep hygiene
- Stress management
- Exposure-based techniques (for specific phobias)
- Avoid caffeine, stimulants, excessive alcohol consumption and diet pills

PHARMACOLOGICAL MEASURES

SSRI
SNRI



SSRI & SNRI

- When discontinuing, taper gradually (especially if used >6 weeks to avoid antidepressant discontinuation syndrome)
- Caution with increased bleeding risk

SSRI	• 1st line for anxiety and related disorders • 1st line in OCD (when drug therapy is needed) • Citalopram and escitalopram are associated with the highest risk of QT prolongation
SNRI	• Venlafaxine: 1st line in all types of anxiety and related disorders, except OCD where it is 2nd line • Duloxetine: 1st line used in GAD only

PHARMACOLOGICAL MEASURES

TCA
MAOI

TCA & MAOI

TCA	• 2nd line in PD, OCD, GAD • 2-3 months needed to see maximal effect • Start low and titrate; discontinue gradually Secondary amines have a lower incidence of side effects compared to tertiary amines like trimipramine, imipramine and clomipramine • Desipramine: Lowest anticholinergic side effects risk • Clomipramine: Highest seizure risk; most serotonergic; favored in OCD • Doxepin: Highest risk of antihistaminergic side effects • Nortriptyline: Lowest risk of hypotension
MAOI	• Used as 2nd line in SAD and PTSD (last line for patients with persistent symptoms) • Dose in the morning to avoid overstimulation and insomnia • Avoid eating tyramine rich foods (irreversible MAOIs: phenelzine, tranylcypromine) • Reversible MAOIs (moclobemide) are better tolerated than irreversible, and are associated with less sexual dysfunction

PHARMACOLOGICAL MEASURES

BENZODIAZEPINES

- Adjunct in early treatment, especially for acute anxiety & agitation
- Provide short term & rapid relief until the other medications can take effect
- Not recommended for long-term use. Risk of abuse, sedation, cognitive impairment, falls and dependence
- Low doses of high-potency benzodiazepines are recommended
- Taper off slowly to prevent withdrawal symptoms
- Avoid consumption of grapefruit with diazepam
- Avoid in: elderly (caution), comorbid SUD

Benzodiazepines are 2nd line for anxiety and related disorders

Potency	Agents	Anxiolytic	Sedating
High	Long acting: Clonazepam	++	+
	Intermediate acting: Lorazepam	+++	++
	Alprazolam	++	+
	Bromazepam	++	+
Medium	Long acting: Diazepam	+++	++

PHARMACOLOGICAL MEASURES

OTHERS

Propranolol	• Used in specific task-related anxiety • Used for tremors and palpitations • Fear of public speaking/stage fright may benefit from low doses taken 30 mins before event • Limited data
Buspirone	• 2nd line in GAD (role in mild-moderate GAD) • Slow onset of effect compared to benzodiazepines • Multiple daily dosing (BID or TID, NOT PRN!) • Less sedating, low addiction tendencies, minimal withdrawal compared to benzodiazepines • Substrate of CYP3A4
Pregabalin	• Monotherapy as first-line for SAD and GAD • Rapid onset of relief (i.e. 1 week) • Also as Adjunctive therapy with partial relief from SSRIs or SNRIs • Requires significantly elevated doses which are poorly tolerated • Side effects: drowsiness: fatigue, impaired coordination, diplopia, and peripheral edema • Avoid CNS depressants

1ST LINE DRUG THERAPIES TO TREAT ANXIETY AND RELATED DISORDERS

	PD	SAD	GAD	OCD	PTSD
Citalopram	✓	X	X	X	X
Escitalopram	✓	✓	✓	✓	X
Fluoxetine	✓	X	X	✓	✓
Fluvoxamine	✓	✓	X	✓	X
Paroxetine	✓	✓	✓	✓	✓
Sertraline	✓	✓	✓	✓	✓
Duloxetine	X	X	✓	X	X
Venlafaxine	✓	✓	✓	X	✓
Pregabalin	X	✓	✓	X	X

Look for trends and outliers! For example, paroxetine & sertraline are first-line for all conditions; venlafaxine is first-line for all EXCEPT OCD

2nd LINE PHARMACOLOGICAL THERAPIES

SSRI Benzodiazepine
SNRI MAOI
TCA Other

Disorder	2 nd line
PD	Imipramine, Clomipramine, Mirtazapine, Alprazolam, Clonazepam, Diazepam, Lorazepam
SAD	Alprazolam, Clonazepam, Bromazepam, Citalopram, Gabapentin, Phenelzine
GAD	Alprazolam, Bromazepam, Lorazepam, Diazepam, Bupropion XL, Buspirone, Hydroxyzine, Imipramine, Quetiapine XR, Vortioxetine
OCD	Citalopram, Clomipramine, Mirtazapine, Venlafaxine XR
PTSD	Fluvoxamine, Mirtazapine, Phenelzine

ADJUSTING THERAPY

Trial 1st line therapy for 6-8 weeks prior to adjusting unless patient has intolerable side effects. Full response can take up to 12 weeks or more.

Improvement

- Continue therapy for 12-24 months
- Monitor to ensure no recurrence of symptoms

Partial Response

- Ensure 1st line agent is at optimal dose, if not, increase the dose
- If at optimal dose, add adjunctive therapy
- Adjunctive agents include anticonvulsants (pregabalin, gabapentin), antipsychotics (e.g. olanzapine, risperidone, quetiapine, aripiprazole)

No response

- If no response at optimal dose, change to another first-line therapy
- Consider a different class if multiple 1st line options fail

PREGNANCY AND BREASTFEEDING

Always use the lowest effective dose

Prior to pregnancy

- Screen for anxiety symptoms (mood and suicidal tendencies)
- Referral to psychiatrist may be necessary if significant distress or interference with functioning



During Pregnancy

- Psychological treatment: **CBT is first-line in mild-moderate anxiety**, is safe without restriction in pregnancy
- Relaxation techniques
- **If severe + situations that affect fetal or maternal safety → Preferred in GAD: escitalopram, sertraline**
- **Avoid Paroxetine: possible CV malformations (3rd line)**
- **Avoid benzodiazepines** (sedation risk and respiratory problems in infants). PRN use may be considered in acute anxiety after weighing risks vs. benefits



Post-partum/ Breastfeeding

- Anxiety can impede mother's ability to care for child and breastfeed
- **Nonpharmacological therapy** preferred
- If drug therapy necessary, → **Preferred in GAD: escitalopram, sertraline**



MONITORING PARAMETERS

Efficacy endpoints			
Parameter	Monitoring Tool	Degree of Change	Timeframe
Symptomatic Anxiety	Patient report GAD-7	Eliminate or decrease	6-8 weeks
Anxiety based disability		Eliminate or decrease	6-8 weeks
Recurrent symptoms		Prevent	Ongoing

Safety Endpoints		
Side effects from medication therapy (sedation, falls, etc.)	Prevent	Duration of therapy

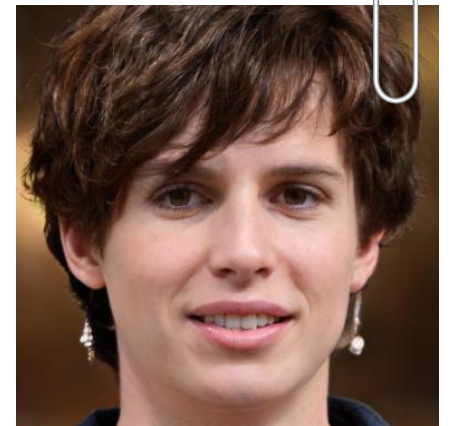
TAKEAWAY

- Anxiety and anxiety-related disorders are common mental health conditions that cause excessive worry, poor concentration, and other symptoms in patients
- It takes longer for relief of symptoms in these disorders than in depression
 - Patients are also on therapy longer – patients should receive treatment for at least 1 year
- First line agents
 - **Panic Disorder**: citalopram, escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline, venlafaxine
 - **Social Anxiety Disorder**: escitalopram, fluvoxamine, paroxetine, sertraline, venlafaxine, pregabalin
 - **Generalized Anxiety Disorder**: escitalopram, paroxetine, sertraline, duloxetine, venlafaxine, pregabalin
 - **Obsessive-Compulsive Disorder**: escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline
 - **Post-Traumatic Stress Disorder**: fluoxetine, paroxetine, sertraline, venlafaxine
- In pregnancy and breastfeeding, non-pharmacologic measures are 1st line (such as psychotherapy)
 - If pharmacotherapy is needed, escitalopram and sertraline are preferred

CASE 1

LE, a 36-year-old female presents to the pharmacy requesting a new rescue inhaler. For the past 2 months, LE has experienced several episodes of SOB, palpitations, dizziness, chest pain and diaphoresis. These episodes have come on unexpectedly and sometimes lead LE to think she was dying. She thinks they are asthma exacerbations, but her salbutamol has had no effect. She often thinks and worries about when another episode will begin and has been using salbutamol prophylactically up to 10 times a day in hopes of preventing this. Because she has a family history of CAD, she also worries this may be a cardiac condition and has been using aspirin prophylactically. She has been calling in sick to work and avoiding going out with friends because she worries about having an episode in public. LE visited her GP recently and was told her asthma is under control. LE has a past medical history of asthma and remote history of opioid abuse (receiving treatment).

She currently takes fluticasone 250 mcg 1 puff BID, salbutamol 2 puffs Q1H PRN, Suboxone® (buprenorphine/naloxone) 8 mg daily, Advil® Cold & Sinus (ibuprofen and pseudoephedrine) Q4H PRN (allergies), and Aspirin® PRN pain (using more recently). She is an architect who lives alone, she has 2 alcoholic drinks per week and 4 cups of coffee per day and she does not smoke. She has no allergies to medications; she just has seasonal allergies. Today her BP is 127/82 mmHg, her HR is 89 bpm, her RR is 15 breaths/min, and her CBC, TSH, and vitamin B12 all appear normal.



CASE 1 - QUESTIONS

1. What is the likely diagnosis for LE?
 - a) Asthma exacerbation
 - b) Agoraphobia
 - c) Acute coronary syndrome
 - d) Panic disorder
2. What are the potential drug causes of this condition?
 - a) Advil Cold & Sinus® (ibuprofen and pseudoephedrine)
 - b) Salbutamol
 - c) Caffeine
 - d) All of the above
3. Which therapy would be best for LE?
 - a) Duloxetine
 - b) Fluvoxamine
 - c) Buspirone
 - d) Diazepam
4. Which of the following is least likely to contribute to LE's condition?
 - a) Caffeine intake
 - b) Salbutamol use
 - c) Aspirin® use
 - d) Asthma

CASE 1 PART 2

LE is a 36-year-old female with a history of asthma. She was recently diagnosed with panic disorder and was started on fluvoxamine 25 mg one week ago. It has now been titrated to 100mg daily. She also started taking loratadine 10mg daily PRN (1-2/week) for seasonal allergies. Today her BP is 127/82 mmHg, her HR is 79 bpm, her RR is 15 breaths/min, and her CBC, TSH, and vitamin B12 all appear normal.



CASE 1 PART 2 - QUESTIONS

1. What is the earliest that LE can expect to see a therapeutic effect with her medication?
 - a) 1 week
 - b) 6 weeks
 - c) 8 weeks
 - d) 12 weeks
2. What is the next best option to recommend for LE?
 - a) Start Cognitive Behavioural Therapy in addition to fluvoxamine
 - b) Stop fluvoxamine and start Cognitive Behavioural Therapy
 - c) Add diazepam
 - d) Add buspirone
3. Clonazepam 0.25 mg PO BID was prescribed for LE. Which of the following is the most appropriate course of action?
 - a) Fill the prescription as is
 - b) Counsel the patient on dependence risk
 - c) Call the prescriber to change the prescription to PRN
 - d) Call the prescriber to recommend a different agent

CASE 2

PA, a 22-year-old male, presents to the pharmacy with a new prescription for lorazepam. For the past 6 months, PA has been more anxious and has recurring thoughts about failing his exams despite being a good student. He finds himself fidgeting with his pen regularly and ensuring that his room is perfectly organized and in the exact orientation he left it. Each time he notices that something isn't perfect with his room, he must organize it or else he won't stop thinking about it. He went to his doctor as this is starting to affect his ability to study. PA is a 4th year university student. He has no significant past medical history and does not take any medications or vitamins. He has an allergy to shellfish. Today, his BP is 112/77 mmHg, his HR is 67 bpm, and his RR is 16 breaths/min.



CASE 2 - QUESTIONS

1. Which of the following conditions does PA most likely have?
 - a) GAD
 - b) OCD
 - c) SAD
 - d) PD
2. Which of the following pharmacological recommendations would be best for PA?
 - a) Ask the doctor to add venlafaxine in addition to lorazepam
 - b) Ask the doctor to switch lorazepam to diazepam for better maintenance effect
 - c) Ask the doctor to switch lorazepam to sertraline
 - d) None required, fill the lorazepam and follow up in 2 weeks
3. Which of the following non-pharmacological measures is least effective as adjunctive management?
 - a) CBT
 - b) Massage therapy
 - c) Aerobic exercise
 - d) Improved sleep hygiene

CASE 3

NN is a 50-year-old male who presents to the pharmacy for advice. For the past 7 months, NN has been having unusual episodes where he suddenly feels scared and anxious. He cannot identify any triggers. These episodes last about 10 minutes, and during this time he feels his heart beating more quickly, feels sweaty, and shakes uncontrollably. He wants to know if taking melatonin could help him with these symptoms. He is the CEO of a chemical factory, he started using cannabis 3 months ago, he stopped using alcohol 2 years ago, and he has been widowed for 5 years. He has a medical history of performance anxiety x 15 years and alcohol use disorder for which he takes propranolol 10mg PRN and naltrexone 100mg daily. Today his BP is 139/88 mmHg, his HR is 99 bpm, and his RR is 20 breaths/min.



CASE 3 - QUESTIONS

1. Which of the following conditions does NN most likely have?
 - a) GAD
 - b) OCD
 - c) SAD
 - d) PD
2. Which of the following recommendations would be best for NN?
 - a) Trial the melatonin immediately when these episodes begin
 - b) Recommend dimenhydrinate PRN instead of melatonin for these episodes
 - c) Advise NN to see his doctor and inform him that melatonin is not the best option
 - d) Recommend increasing his cannabis use to reduce anxiety
3. Which of the following medications would be least effective for NN?
 - a) Mirtazapine
 - b) Citalopram
 - c) Venlafaxine
 - d) Paroxetine

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CHANGE LOG

August 2025

- Minor formatting and phrasing changes throughout
- Slide 3: Added SUD to legend
- Slide 13: Updated buspirone from TID to BID or TID, added it's not PRN
- Slide 14: Updated lists to 1st line chart for clarity/better focus
- Slide 17: Added takeaway
- Slide 19: Updated choices for question 3
- Slide 21: Updated choices for question 2 and changed question 3

November 2025

- Slide 6: Expansion of definition to include excessive
- Slide 9: Simplified definition of CBT for all anxieties
- Slide 12: Simplification of populations to avoid
- Slide 15: Added 2nd line therapies
- Slide 16: Added more details on benzodiazepine use in pregnancy

March 2026

- Slide 7: Including monitoring tools for anxiety
- Slide 17: Moved info about what to do if 1st line is ineffective to its own slide
- Slide 19: Included how to monitor efficacy